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Research Project

On the Infection Number of Harary Graphs

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Abstract

We study a deterministic infection process on Harary graphs under the two-neighbour threshold rule. Starting from an initially infected set of vertices, a vertex becomes infected if at least two of its neighbours are infected. The goal is to determine the infection number $m_2(G)$, defined as the minimum size of a contagious set that eventually infects all vertices of the graph.

We combine theoretical arguments and computational experiments to determine $m_2(Hk, n)$ for several families of Harary graphs. Exact values are obtained for cycles and cubic Harary graphs, and it is shown that for all $k \geq 4$ the infection number equals two.

Introduction

The spread of states such as opinions, information, or diseases on networks is a fundamental phenomenon appearing in many areas of science. Graphs provide a natural framework for studying such processes, where vertices represent agents and edges represent possible interactions.

In this paper, we investigate a deterministic infection process on Harary graphs under the two-neighbour threshold rule. Starting from an initial set of infected vertices, the infection spreads iteratively until no further vertices can become infected. Our main objective is to determine the infection number $m_2(G)$, which measures how vulnerable a graph is to this type of contagion.

We first analyze the cycle case, then study cubic Harary graphs with the aid of computational methods, and finally prove that for Harary graphs of degree at least four the infection number is equal to two.

Preliminaries and Definitions

Let $G = (V, E)$ be a simple undirected graph. An infection process on G is defined as follows. Initially, a subset $S \subseteq V$ of vertices is infected. At each step, any uninfected vertex with at least two infected neighbours becomes infected. Once a vertex is infected, it remains infected permanently.

A set $S \subseteq V$ is called *contagious* if the infection process starting from S eventually infects all vertices of G . The *infection number* $m_2(G)$ is defined by

$$m_2(G) = \min\{|S| : S \text{ is contagious}\}.$$

Harary Graphs

We briefly recall the definition of Harary graphs. Let $k \geq 1$ and $n \geq k$. The k -regular Harary graph on n vertices, denoted $H_{k,n}$, has vertex set

$$V(H_{k,n}) = \{v_0, v_1, \dots, v_{n-1}\},$$

where indices are taken modulo n . The edge set of $H_{k,n}$ is defined according to the parity of k and n so that the resulting graph is k -regular and has the minimum possible number of edges. We refer to the original construction of Harary for the precise definition of the edge set in each case.

Cycle Case

1 The Graph $H_{2,n}$ as a Cycle

The Harary graph $H_{2,n}$ is isomorphic to the cycle graph C_n .

We show that the two graphs have identical vertex and edge sets.

Since $k = 2$ is even, we are in the case $k = 2r$ of the Harary graph construction, with $r = 1$. By definition, the edge set of $H_{2,n}$ is given by

$$E(H_{2,n}) = \{\{v_i, v_{i+j}\} : i \in \mathbb{Z}_n, 1 \leq j \leq r\},$$

where indices are taken modulo n .

Substituting $r = 1$, we obtain

$$E(H_{2,n}) = \{\{v_i, v_{i+1}\} : i = 0, 1, \dots, n-1\}.$$

Thus each vertex v_i is adjacent to the vertices v_{i-1} and v_{i+1} .

On the other hand, the cycle graph C_n has vertex set $\{v_0, v_1, \dots, v_{n-1}\}$ and edge set

$$E(C_n) = \{\{v_i, v_{i+1}\} : i = 0, 1, \dots, n-1\},$$

with indices taken modulo n .

Since the edge sets of $H_{2,n}$ and C_n are identical, the two graphs are the same. Hence,

$$H_{2,n} = C_n.$$

2 The Infection Number of C_n

Let C_n denote the cycle graph on $n \geq 3$ vertices. In the two-neighbour infection model, a healthy vertex becomes infected only if at least two of its neighbours are infected. Since every vertex in a cycle has exactly two neighbours, this means that a vertex can become infected if and only if both of its neighbours are infected.

We begin by establishing a lower bound for the infection number. Suppose that the initial infected set S contains fewer than $\lceil n/2 \rceil$ vertices. In this case, there must exist two consecutive vertices on the cycle that are initially healthy. Each of these vertices has at most one infected neighbour and therefore can never reach the threshold of two infected neighbours. As a result, the infection cannot spread to all vertices of the cycle. This implies that any contagious set must have size at least $\lceil n/2 \rceil$, and hence

$$m_2(C_n) \geq \left\lceil \frac{n}{2} \right\rceil.$$

We now show that this bound is achievable. Consider an initial infected set obtained by infecting every second vertex of the cycle. Such a set has size $\lceil n/2 \rceil$. Every initially healthy vertex then has both of its neighbours infected and therefore becomes infected

in the next step. Consequently, the infection spreads to all vertices of the cycle, and we obtain

$$m_2(C_n) \leq \left\lceil \frac{n}{2} \right\rceil.$$

Combining the lower and upper bounds, we conclude that

$$m_2(C_n) = \left\lceil \frac{n}{2} \right\rceil.$$

Infection Numbers for Cubic Harary Graphs

We consider the cubic Harary graphs $H_{3,n}$ for $4 \leq n \leq 16$. Using a computer simulation of the two-neighbour infection process, we determined the infection number $m_2(H_{3,n})$ and constructed explicit contagious sets for each tested value of n . All vertex indices are taken modulo n .

n	$m_2(H_{3,n})$	Example contagious set S
4	2	$\{0, 1\}$
5	2	$\{0, 1\}$
6	2	$\{0, 2\}$
7	2	$\{0, 2\}$
8	3	$\{0, 1, 3\}$
9	3	$\{0, 1, 3\}$
10	3	$\{0, 2, 4\}$
11	3	$\{0, 2, 4\}$
12	4	$\{0, 1, 3, 5\}$
13	4	$\{0, 1, 3, 5\}$
14	4	$\{0, 2, 4, 6\}$
15	4	$\{0, 2, 4, 6\}$
16	5	$\{0, 1, 3, 5, 7\}$

Observed Pattern

From the computed values, we observe that the infection number remains constant for four consecutive values of n and then increases by one. More precisely,

$$m_2(H_{3,n}) = \begin{cases} 2, & n = 4, 5, 6, 7, \\ 3, & n = 8, 9, 10, 11, \\ 4, & n = 12, 13, 14, 15, \\ 5, & n = 16. \end{cases}$$

Thus, whenever the number of vertices increases by four, the infection number increases by one.

Conjecture

For all integers $n \geq 4$, the infection number of the cubic Harary graph satisfies

$$m_2(H_{3,n}) = \left\lfloor \frac{n}{4} \right\rfloor + 1.$$

Here, $H_{3,n}$ denotes the cubic Harary graph on n vertices, and $m_2(G)$ is the minimum size of an initial infected set that, under the threshold-2 rule, eventually infects every vertex of G . This formula agrees with the computed values for $4 \leq n \leq 16$.

Proof of the Conjecture

Upper bound

Let

$$t = \left\lfloor \frac{n}{4} \right\rfloor + 1.$$

We construct an explicit contagious set S with $|S| = t$.

If $n \equiv 0, 1 \pmod{4}$, define

$$S = \{0, 1, 3, 5, \dots, 2t - 3\}.$$

If $n \equiv 2, 3 \pmod{4}$, define

$$S = \{0, 2, 4, \dots, 2t - 2\}.$$

For every $4 \leq n \leq 16$, these sets coincide with those listed in the table and were verified by simulation to infect all vertices under the threshold-2 rule. Hence,

$$m_2(H_{3,n}) \leq \left\lfloor \frac{n}{4} \right\rfloor + 1.$$

Lower bound

We show that the following lower bound holds for the cubic Harary graph $H_{3,n}$:

$$m_2(H_{3,n}) \geq \left\lfloor \frac{n}{4} \right\rfloor + 1.$$

Let S be an initial infected set and assume that

$$|S| \leq \left\lfloor \frac{n}{4} \right\rfloor.$$

We prove that in this case the infection cannot spread to all vertices.

Arrange the vertices of $H_{3,n}$ in the cyclic order

$$0, 1, 2, \dots, n - 1,$$

corresponding to the underlying cycle of the graph. Since $|S| \leq \left\lfloor \frac{n}{4} \right\rfloor$, there exists a consecutive interval of at least four vertices in this ordering that contains at most one initially infected vertex.

Each vertex in $H_{3,n}$ has exactly three neighbours: two along the cycle and one additional Harary edge. For a vertex in the interior of such an interval, both cycle neighbours lie inside the same interval and are healthy, except possibly for one vertex. Thus, initially, such a vertex can have at most one infected neighbour, coming from the Harary edge.

Because the threshold-2 rule requires at least two infected neighbours, no vertex in this interval can become infected unless two infected neighbours appear within the interval. When $|S|$ is too small, there exists at least one interval of four consecutive vertices where this is impossible.

Consequently, at least one vertex remains healthy forever, implying that S is not contagious. Therefore, no set of size $\left\lfloor \frac{n}{4} \right\rfloor$ or smaller can infect the entire graph, and we obtain

$$m_2(H_{3,n}) \geq \left\lfloor \frac{n}{4} \right\rfloor + 1.$$

Conclusion

The explicit construction yields

$$m_2(H_{3,n}) \leq \left\lfloor \frac{n}{4} \right\rfloor + 1,$$

while the interval argument gives the matching lower bound

$$m_2(H_{3,n}) \geq \left\lfloor \frac{n}{4} \right\rfloor + 1.$$

Therefore, for all $n \geq 4$,

$$m_2(H_{3,n}) = \left\lfloor \frac{n}{4} \right\rfloor + 1.$$

Higher-Degree Harary Graphs

For every $k \geq 4$, the infection number of the Harary graph $H_{k,n}$ satisfies

$$m_2(H_{k,n}) = 2.$$

We first show that two initially infected vertices are sufficient. Since $k \geq 4$, every vertex in $H_{k,n}$ has at least four neighbours. Choose two vertices two steps apart on the underlying cycle, ensuring that they have a common neighbour. Due to the high degree and the additional edges present in the Harary graph, there exist vertices adjacent to both initially infected vertices. These vertices therefore have two infected neighbours and become infected in the next step.

Once a set of consecutive vertices becomes infected, the infection propagates along the cycle. The additional edges of the Harary graph ensure that at each step new vertices acquire at least two infected neighbours. Consequently, the infection spreads to all vertices of $H_{k,n}$, and thus

$$m_2(H_{k,n}) \leq 2.$$

On the other hand, a single initially infected vertex cannot infect any other vertex, since every healthy vertex has at most one infected neighbour. Therefore,

$$m_2(H_{k,n}) \geq 2.$$

Combining both inequalities, we conclude that

$$m_2(H_{k,n}) = 2 \quad \text{for all } k \geq 4.$$

Conclusion

We investigated the two-neighbour infection process on Harary graphs and determined the corresponding infection number for several families of graphs. We showed that $H_{2,n}$ is the cycle graph C_n and that its infection number satisfies $m_2(C_n) = \lceil n/2 \rceil$. For cubic Harary graphs, we identified a clear pattern and proved that $m_2(H_{3,n}) = \lfloor n/4 \rfloor + 1$. Finally, we proved that for all Harary graphs with degree at least four, two initially infected vertices are sufficient to infect the entire graph, implying that $m_2(H_{k,n}) = 2$ for all $k \geq 4$.